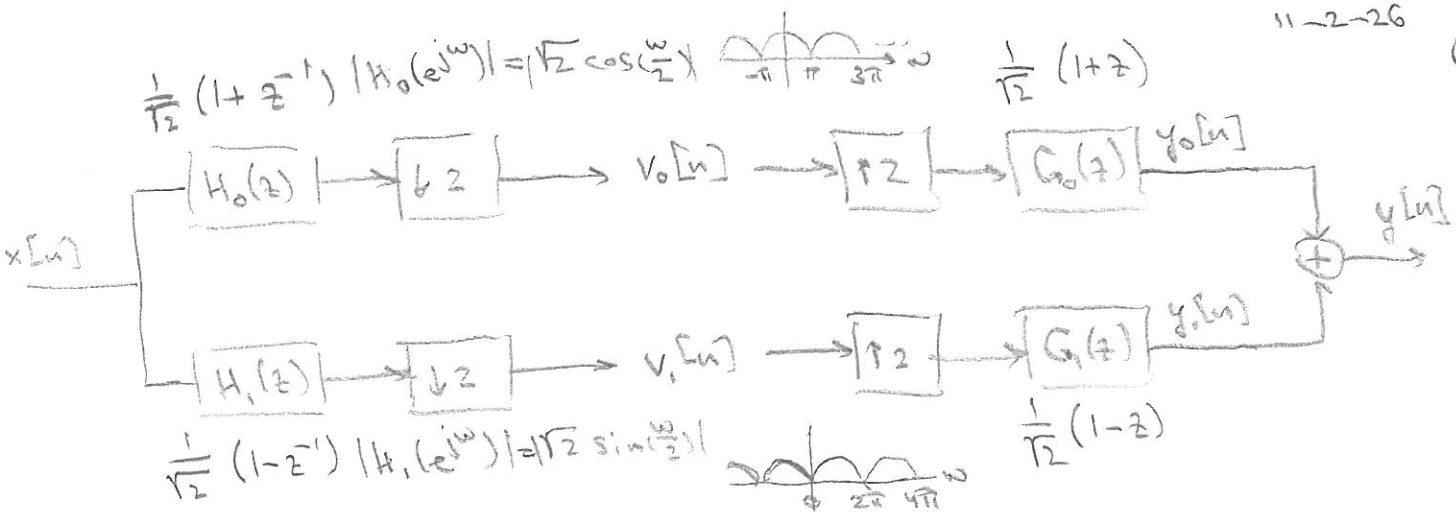


$$Y(z) = \frac{1}{2}\left[G_0(z)H_0(z) + G_1(z)H_1(z)\right]X(z) + \frac{1}{2}\left[G_0(z)H_0(-z) + G_1(z)H_1(-z)\right]X(-z)$$

$$G_0(z)H_0(z) + G_1(z)H_1(z) = \frac{1}{2}\left[(1+z)(1+z^{-1}) + (1-z)(1-z^{-1})\right]$$

$$= \frac{1}{2}\left[1 + z + z^{-1} + 1 + 1 - z - z^{-1} + 1\right] = 2$$

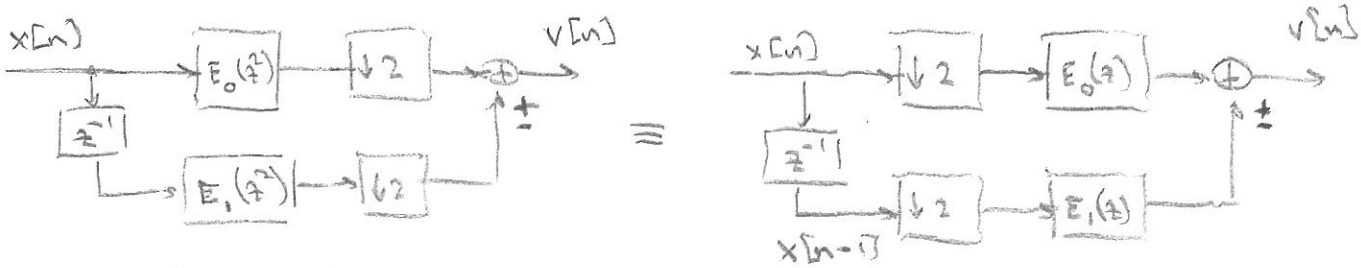
$$G_0(z)H_0(-z) + G_1(z)H_1(-z) = \frac{1}{2}\left[(1+z)(1-z^{-1}) + (1-z)(1+z^{-1})\right]$$

$$= \frac{1}{2}\left[1 + z - z^{-1} - 1 + 1 - z + z^{-1} - 1\right] = 0$$

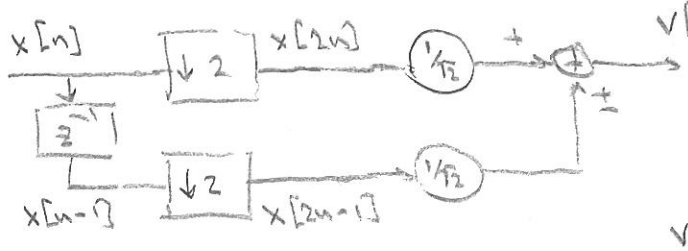
$$H(z) = \frac{1}{\sqrt{2}}(1 \pm z^{-1})$$

$$e_0[n] = h[2n] = \frac{1}{\sqrt{2}} \qquad E_0(z) = \frac{1}{\sqrt{2}} = R_0(z^2)$$

$$e_1[n] = h[2n+1] = \pm\frac{1}{\sqrt{2}} \qquad E_1(z) = \pm\frac{1}{\sqrt{2}} = R_1(z^2)$$



For $H(z) = \frac{1}{\sqrt{2}}(1 \pm z^{-1})$

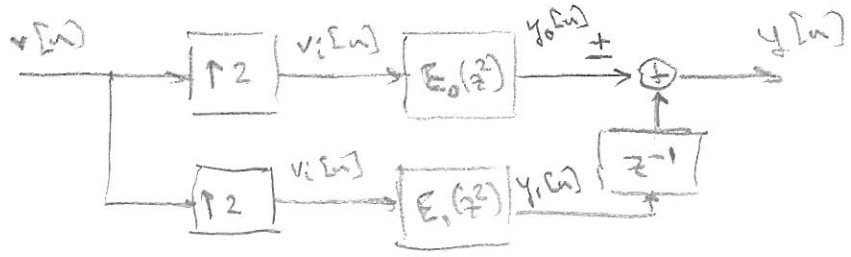


$$v[n] = \frac{1}{\sqrt{2}}\left[x[2n] \pm x[2n-1]\right]$$

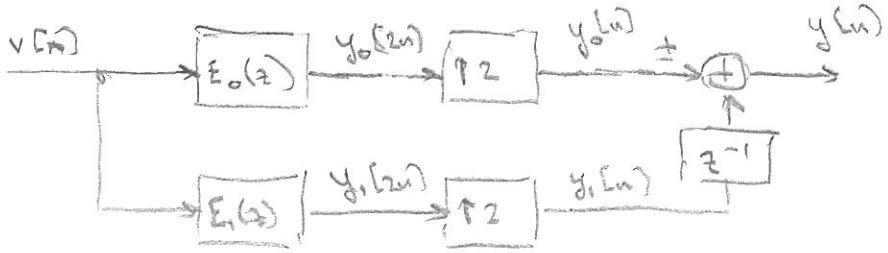
$$v_0[n] = \frac{1}{\sqrt{2}}\left[x[2n] + x[2n-1]\right]$$

$$v_1[n] = \frac{1}{\sqrt{2}}\left[x[2n] - x[2n-1]\right]$$

$$z^{-1}G_0(z) = \frac{1}{\sqrt{2}}(1 \pm z)z^{-1} = \frac{1}{\sqrt{2}}(z^{-1} \pm 1)$$

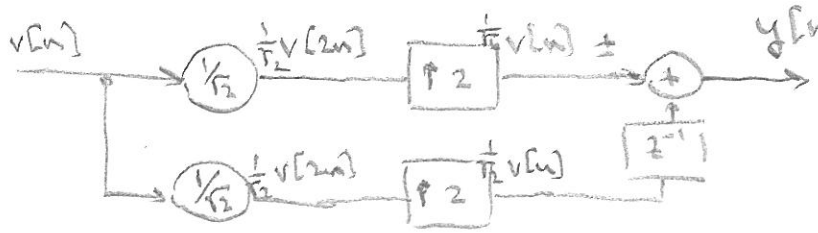


≡



For $z^{-1}G_0(z) = \frac{1}{\sqrt{2}}(z^{-1} \pm 1)$ $\quad E_0(z) = \pm\frac{1}{\sqrt{2}}$

$E_1(z) = \frac{1}{\sqrt{2}}$



$$y[n] = \frac{1}{\sqrt{2}}\left[\pm v[n] + v[n-1]\right]$$

$$y_0[n] = \frac{1}{\sqrt{2}}\left[v_0[n] + v_0[n-1]\right]$$

$$y_1[n] = \frac{-1}{\sqrt{2}}\left[v_1[n] - v_1[n-1]\right]$$

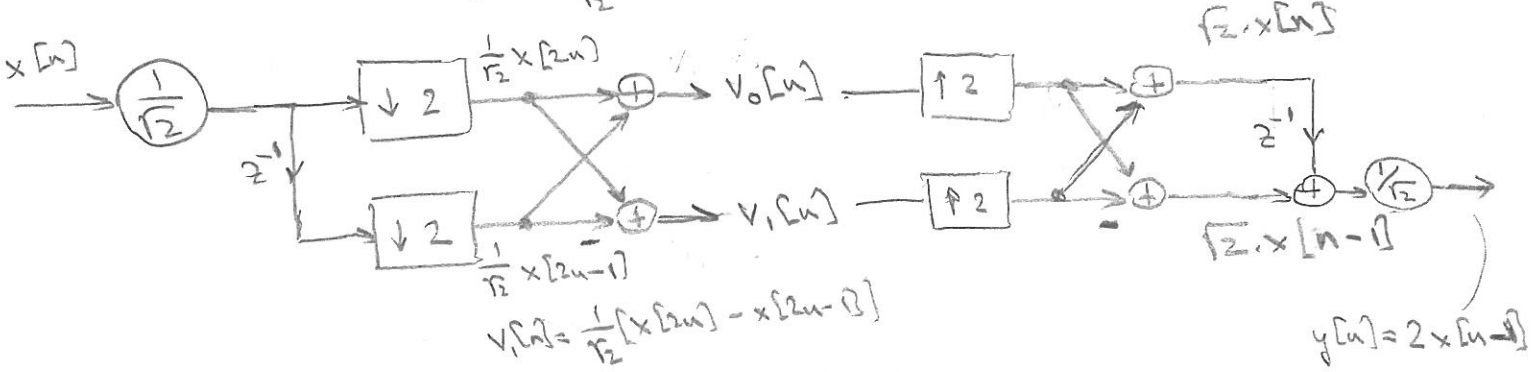
$$y[n] = \frac{1}{\sqrt{2}}\left[v_0[n] + v_0[n-1] - v_1[n] + v_1[n-1]\right]$$

$$= \frac{1}{\sqrt{2}}\left[\frac{1}{\sqrt{2}}\left[x[2n] + x[2n-1] + x[2n-1] + x[2n-2]\right] + \right.$$

$$\left. -\frac{1}{\sqrt{2}}\left[x[2n] - x[2n-1] - x[2n-1] + x[2n-2]\right]\right]$$

$$= \frac{1}{2}\left[4x[2n-1]\right] = 2x[2n-1]$$

$$v_0[n] = \frac{1}{\sqrt{2}}[x[2n] + x[2n-1]]$$



$$v_1[n] = \frac{1}{\sqrt{2}}[x[2n] - x[2n-1]]$$